

Rohan Singh

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EDUCATION

University of Virginia Charlottesville, VA
BS in Computer Science – Minor in Data Science May 2027
• Coursework: Data Structures & Algorithms, Software Engineering, Database Systems, Computer Systems & Organization

EXPERIENCE

Expedia Group Jun 2026 – Present
Machine Learning Engineer Intern / Python, FastAPI, Trino SQL San Jose, CA

- Prototyped a **FastAPI** service for OneKey, a **100M+ member rewards program**, with live profiles and safe fallbacks.
- Measured a causal **+0.9pp booking lift** (placebo-clean, **321k**-member holdout), replicated within **0.7%** out of cohort.
- Showed reward spend was anti-correlated with lift-per-dollar ($\rho = -0.88$); **44%** of budget hit the least-efficient cell.
- Defined a **27-field train/serve contract** from **112** model features and verified **111M/day** trip-intent backfill end to end.
- Hosted the rewards recommender behind the service and ran the full live-features-to-safe-decision path end to end.

Freddie Mac May 2025 – Jul 2025
Software Engineer Intern / Java, TypeScript, Spring Boot, Angular McLean, VA

- Implemented **Java Spring Boot** services to validate 120k accounting events and deployed them through **Jenkins CI/CD**.
- Consolidated 200+ REST endpoints behind **GraphQL**, cutting Angular payloads **60%** and simplifying client integration.
- Benchmarked Playwright against Cypress, cutting test time **4x from 8 to 2 minutes** and driving the team's migration.
- Automated nightly UAT refreshes with Python and Jenkins, cutting load time from **2 hours to 10 minutes per run**.

Biocomplexity Institute Nov 2024 – May 2025
AI/ML Research Assistant / Python, Slurm Charlottesville, VA

- Developed Python graph-analysis pipelines with **cuGraph**, SciPy, and NetworkX over 1.2M-edge financial networks.
- Raised AUC-ROC from 0.55 to **0.76**, surfacing **250 anomalies at 89% precision** for downstream investigator review.
- Scaled end-to-end GPU anomaly detection with **cuGraph** to **2M+ nodes and edges** on UVA's Slurm cluster.

Cubic Corporation Jun 2024 – Aug 2024
Software Engineer Intern / JavaScript, React, Terraform Sterling, VA

- Launched an AWS-hosted React/Docusaurus developer portal on Terraform-managed infra, serving **500+ engineers**.
- Cut onboarding tickets **85%, from 90 to 13 in one month**, by redesigning site flow and navigation after user tests.
- Added Algolia DocSearch and simplified navigation, reaching **95% top-3 accuracy at 120ms median query latency**.
- Integrated AWS CodeBuild lint and test checks on pull requests, reducing manual QA and speeding team merges.

PROJECTS

Shortlist | Go, Python, PyTorch, Next.js
• Served exact retrieval across **87,585 movies** in Go with float16 embeddings and history filtering at **4.0ms median**.
• Raised **HitRate@10 from 73.8% to 84.1%** by aligning training with the product's 1–5 favorite onboarding flow.
• Shipped a Next.js app on Vercel with multi-title onboarding, natural reasons, saved lists, and **Playwright/Axe CI**.

Marathon Prep Bot | TypeScript, Node.js, Garmin FIT SDK
• Wrote a strict TypeScript pipeline that decodes binary Garmin FIT files into **1,378 CRC-validated activities**.
• Automated 6-hour iCloud syncs with **241 regression tests across 32 suites**, keeping GPS and health data local.
• Shipped a live dashboard turning 44 months of activity data into weekly plans, training-load trends, and daily guidance.

Today | Swift, SwiftUI, HealthKit, WorkoutKit
• Built **Today**, a SwiftUI fitness app joining HealthKit data, live coach plans, and a **255-exercise** hand-mapped lifting log.
• Designed offline-first **AES-256-GCM** sync with Keychain credentials, protected backups, retries, and consent-scoped data.
• Integrated WorkoutKit and App Intents for Apple Watch plans and Siri logging; covered security and UI with **217 tests**.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, Go, C++, SQL
Frameworks: Spring Boot, Node.js, Express, React, Next.js, FastAPI, GraphQL, PyTorch
Infrastructure & Data: AWS, GCP, Docker, Terraform, Jenkins, GitHub Actions, PostgreSQL, MySQL, MongoDB